Class-wise Graph Embedding-Based Active Learning for Hyperspectral Image Classification

Xiaolong Liao, *Graduate Student Member, IEEE*, Bing Tu, *Senior Member, IEEE*, Jun Li, *Fellow, IEEE*, and Antonio Plaza, *Fellow, IEEE*

***Abstract*— Deep learning (DL) techniques have shown remarkable progress in remotely sensed hyperspectral image (HSI) classification tasks. The performance of DL-based models highly relies on the quality and quantity of labeled data. However, manual labeling is a laborious and expensive process that requires substantial efforts from human experts. Active learning (AL) techniques have been developed to alleviate the burden of manual annotation by selecting the most informative and uncertain samples for labeling. In this article, we propose a new class-wise graph-embedding-based AL (CGE-AL) framework implemented by a class-wise graph convolutional network (CGCN). First, we train a classifier with labeled data and infer latent features from labeled and unlabeled samples with the trained parameter. Then, we group the labeled data into multiple one-label sets by category. In a class-wise manner, we initialize the nodes of the graph with one-label and unlabeled features, which are then fed into CGCN. By updating the graph parameters with binary loss, CGCNs measure the uncertainty between labeled nodes and unlabeled nodes. To select the most valuable sample for labeling, we adopt the class minimum uncertainty to query the unlabeled nodes with higher overall uncertainty. We repeat this process with the updated labeled set to retrain our classification model and CGCNs. Extensive experiments demonstrate the outstanding performance of our method compared with other state-of-the-art AL-based approaches.**

***Index Terms*— Active learning (AL), deep learning (DL), graph convolutional network (GCN), hyperspectral image (HSI) classification.**

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Xiaolong Liao is with the School of Information Science and Engineering, Hunan Institute of Science and Technology, Yueyang 414000, China (e-mail: xiaolongliao.china@gmail.com).

Bing Tu is with the Institute of Optics and Electronics, the Jiangsu Key Laboratory for Optoelectronic Detection of Atmosphere and Ocean, and the Jiangsu International Joint Laboratory on Meteorological Photonics and Optoelectronic Detection, Nanjing University of Information Science and Technology, Nanjing, Jiangsu 210044, China (e-mail: tubing@nuist.edu.cn).

Jun Li is with the Faculty of Computer Science, China University of Geosciences, Wuhan 430074, China (e-mail: lijuncug@cug.edu.cn).

Antonio Plaza is with the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, Escuela Politecnica, University of Extremadura, E-10003 Cáceres, Spain (e-mail: aplaza@unex.es).

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I. Introduction

HYPERSPECTRAL images (HSIs) are comprised by hundreds of continuous channels and contain a wealth of discriminative spectral information. In recent years, HSIs have attracted increasing interest in the remote sensing community and gained a broad range of applications in several challenging scenarios, such as image reconstruction [1], [2], target detection [3], [4], and medical diagnosis [5], [6]. As one of the core tasks of HSI analysis, HSI classification assigns labels to pixel-wise objects by identifying their spectral characteristics and spatial structural information [7], [8].

Recently, many advanced deep learning (DL) frameworks have been developed for HSI classification tasks, such as convolutional neural networks (CNNs) [9], [10], graph convolutional networks (GCNs) [11], [12], and vision transformers (ViTs) [13], [14]. These methods have proven to be capable of automatically discovering valuable representations from large amounts of annotated data. However, high-quality data annotation is a time-consuming and expensive task in remote sensing committee. Moreover, spectral similarities and intra-class spectral variabilities [15] make accurately classification more difficult as the training set expands. There is a gap in the optimization of deep neural network architectures with respect to the representativeness of the training data. To address these issues, a considerable amount of researches have been conducted on how to better exploit limited supervised information [16], [17], [18].

Active learning (AL) has been widely investigated to actively select the most valuable samples for labeling to boost the predict performance of the model. In the AL mode, a small subset of reference data is randomly selected as the labeled set to train a classification model. This model is then used to determine which unlabeled samples would have the best performance. In the remote sensing community, AL techniques have been applied in HSI classification tasks to alleviate the burden of hyperspectral data annotation. These approaches can be roundly categorized into three major groups: *committee*, *representativeness*, and *uncertainty*-based AL. The committee-based methods query the maximum divergences of the unlabeled samples between multiview or multiscale models. The representativeness-based methods seek to match the unlabeled data distribution with the labeled one. The uncertainty-based approaches label the samples that are most ambiguous for the current model.

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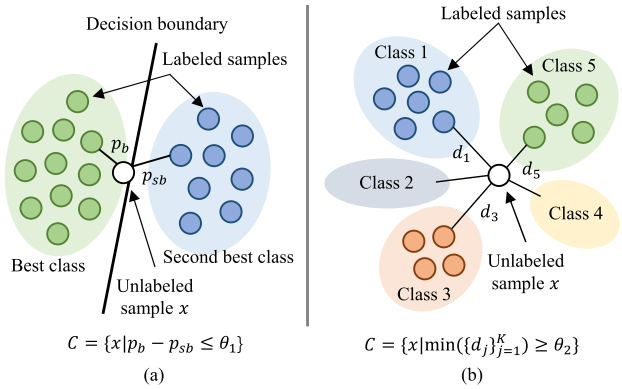


Fig. 1. Toy illustrates the comparison of two kinds of sampling strategies. (a) BvSB queries the predict probabilities p_b and p_{sb} of the unlabeled sample with a threshold. (b) Our method uses a stricter constraint that ensures the entire uncertainty of all the classes surpasses the threshold to guarantee the unlabeled data sufficient distinct from the labeled data.

More recently, researchers have attempted to integrate AL strategies with DL architectures to alleviate the reliance on annotated data in the model optimization process. To name a few, Bayesian CNNs have been developed with different architectures and sampling techniques to improve the classification performance in [19]. Yao et al. [20] used entropy to measure the label consistency within superpixels and adopted the best-vs-second best (BvSB) strategy to query the samples in a semi-active CNN framework. Under a prototypical network architecture, a semi-supervised clustering method was used in [21] by computing the distance of samples to the mean vectors of all the classes to query the informative samples. Wang et al. [22] incorporated both the representative and informative criteria (with adversarial AEs) by adopting a multivariate and distributed distance scheme.

Although these methods are effective in selecting valuable unlabeled data, they also exhibit two limitations. First, in some of the existing approaches, the sampling function used the model result (i.e., class probabilities) to query the diverse and uncertain samples. Consequently, the quality of the selected data heavily relies on the confidence of the classifier's prediction results. Second, as depicted in Fig. 1(a), the BvSB strategy measures the uncertainty of unlabeled data by computing the margin between the probabilities of the largest and second largest classes, disregarding the other classes. This approach captures the distinction between these two classes, but may overlook the overall uncertainty distribution across all the classes.

To address the above issues, we propose a graph embedding and uncertainty-based AL method, called class-wise graph embedding-based AL (CGE-AL). Fig. 1(b) illustrates the concept of our uncertainty sampling model, which queries the class minimum uncertainty by our sampling function to guarantee the entire uncertainty distribution of unlabeled data surpasses a threshold. First, a small amount of reference data are randomly selected as the labeled set to train the classifier. The latent features of labeled and unlabeled samples are extracted with the trained parameter. Based on the true labels, the labeled set is split into multiple one-label sets by category. Then, we build one-label graphs using the features of one-label and unlabeled data for each category. After the graphs pass through class-wise GCN (CGCN), the unlabeled nodes with high class minimum uncertainty are selected with a predefined threshold. Following, k-means is performed on their feature representations to select the most representative samples. The queried samples are then labeled and fed into the labeled set to update the classifier and CGCNs in the next stage.

The main contributions of the proposed method can be summarized as follows.

1) A new uncertainty-based acquisition function is proposed that is independent of the machine learning model. It inherits multiclass uncertainty from the model while producing an extra binary uncertainty for sampling in the graph embedding space.
2) The CGCN architecture is designed to measure the relationship of unlabeled samples to labeled sample from each class. This strategy removes the negative effect from other unrelated classes and avoids redundant graph connections in sampling stage.
3) The comparative experiments, including three benchmark HSI datasets and a new satellite application, demonstrate that our method outperforms several state-of-the-art AL methods and sampling techniques in terms of accuracy, visualization, and sampling behavior.

The remainder of this article can be summarized as follows. Section II summarizes previous work on uncertainty-based and graph-based AL strategies. Section III introduces the proposed CGE-AL method, focusing on model training, uncertainty sampling, and candidate selection. Sections IV and V illustrates the parameter settings and experimental results with respect to other state-of-the-art approaches and also provides some discussions. Section VI concludes the article with some remarks and hints at plausible future research lines.

II. Related Work

A. Conventional AL With HSI Classification

In the remote sensing community, pixel-wise data annotation is a time-consuming task as it might require field measurements or tedious photo interpretation. To reduce the burden of data annotation, AL techniques have been extensively studied in HSI classification tasks to select the most valuable samples for labeling. The proposed AL methods can be summarized in three groups as follows.

1) Committee-Based AL: Committee-based methods select the samples by querying the maximum divergences between a committee of models [23], [24]. This follows the fact that independent models diverge in the predicted results of unknown data. For instance, the disagreement analysis on multiple views, i.e., subpixel level, pixel level, and superpixel level, is conducted to select diverse samples based on their prediction results [16]. A leave-one-class-out strategy is designed in [17] to overcome the overconfidence of the current model by constructing a set of classifiers, which lack one-class labeled samples to train these classifiers.

2) Representativeness-Based AL: To fit the data distribution of unlabeled set, representativeness-based approaches query representative samples to enrich the labeled set [25], [26]. A modified cluster-based method [27] is used to select samples for manual labeling, which makes the labeled samples

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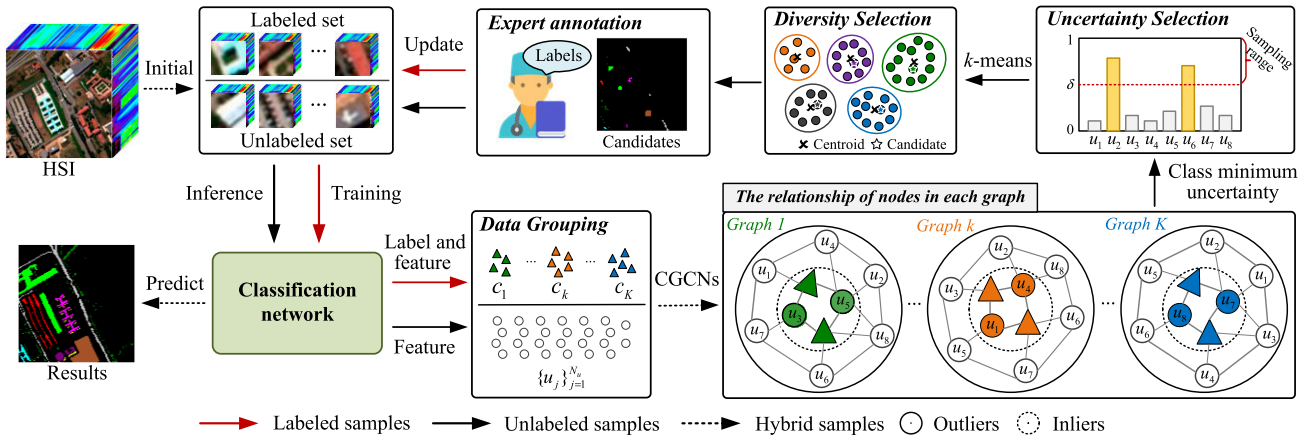


Fig. 2. Overall architecture of our framework. Following the training of classification backbone, labeled data are grouped into K one-label sets based on true labels. Then, CGCN performed on one-label and unlabeled data for each class. The unlabeled node with high confidence is regarded as "inlier." Based on this assumption, nodes that are "outliers" in all the graphs have lower confidence and are selected as uncertain samples. Finally, the selected samples are clustered into B groups to choose the candidates closest to each centroid.

uniformly distributed throughout the data space. Meanwhile, Wang and Ren [28] try to fool a discriminator that distinguishes unlabeled data from labeled data based on a generative adversarial network (GAN), which implicitly bridges the gap between the labeled and unlabeled data distributions.

3) Uncertainty-Based AL: Based on the predictions of unlabeled data, uncertainty-based approaches select the uncertain samples that are the most ambiguous for the present model. They be classified into interclass uncertainty criteria and epistemic uncertainty criteria [29]. Among them, the interclass uncertainty strategy seeks samples lying on the class boundary. For instance, Li et al. [30], [31] introduced entropy-based and breaking-tie-based criterion into machine learning models. Under a one-against-all architecture, Shi et al. [32] select the most uncertain samples using the margin distance to the hyperplanes of a binary SVM classifier. Meanwhile, epistemic uncertainty criteria have been evaluated through uncertainty exploration of DL models. A Bayesian CNN framework is proposed in [19], which offers robustness to overfitting on small labeled sets.

B. Graph-Based AL

The above AL methods are usual designed for independent and identically distributed labeled and unlabeled data, where the sampling functions query each unlabeled sample independently. In the graph structure, the labeled nodes can improve the predictions of the neighbor unlabeled nodes by passing the information between the nodes [33], [34]. Thus, the acquisition function of graph-based AL methods can take the relationship of labeled and unlabeled nodes into consideration.

Recently, with the promising outcome of graph neural networks (GNNs) and GCN architectures in graph learning, some researchers started to investigate whether AL can be applied in training them. For instance, Cai et al. [35] combined the graphical information and node centrality to query the most informative and uncertain nodes for labeling. Wu et al. [36] proposed a GNN-based AL framework that used node feature propagation followed by k-Medoids clustering. Wang et al. [37] proposed an AL framework of GCN-based zero-shot learning by extending the k-center algorithm with a strategy for selecting the most crucial class. CoreGCN [38] measured the relationships of labeled and unlabeled data to seek the uncertain samples. In the HSI classification community, Sun et al. [39] integrated AL and hypergraph embedding using the margin probability to represent interclass correlation. Ye et al. [40] fused local and global features in the GCN framework [38] to gauge the discriminative information of unlabeled data. However, graph embedding serves as a graphical information learning module instead of a sampling module, which selects the informative samples using these methods. Moreover, redundant connections of unrelated classes are not considered in these approaches, which can complicate subsequent analysis and sampling.

III. Proposed Approach

Fig. 2 illustrates the overall architecture of the proposed method. First, a fixed number of labeled and unlabeled sets are selected from reference data. Then, a DL-based classification model is trained using the labeled set to extract the feature representations of samples. Based on their true labels and representations, one-label set and unlabeled sets are used to construct one-label graphs and adjacency matrices for all the categories. Next, the CGCN is used to measure the uncertainty of nodes to classes. Based on the class minimum uncertainty of CGCN's outputs, we queried the uncertain samples and selected the centroids as members of a candidate set by clustering them onto a fixed number of clusters. Finally, the candidate set is annotated by the oracle to retrain the model and CGCN in the next stage. At the end, an extra test set is used to evaluate the classification performance of the model.

A. Problem Definition

In general, the task of AL in HSI classification is designed to train an optimal classification model using a fixed number of labeled samples. At the start of the AL process, a small subset of the reference data is randomly selected with K classes as an initial labeled set $\mathcal{D}_l = \{\boldsymbol{x}_i, y_i\}_{i=1}^{N_l}$, where $\boldsymbol{x}_i$ represents a labeled sample (i.e., a pixel in the HSI), $y_i \in \{1, \ldots, K\}$ denotes its class label, and N_l is the number of labeled samples. At each AL stage, the classification model is trained

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with $\mathcal{D}_l$. Then, the acquisition function actively selects B samples from an unlabeled set $\mathcal{D}_u = \{\boldsymbol{x}_j\}_{j=1}^{N_u}$ by adopting the sampling criteria. Following this, the queried set $\mathcal{D}_q$ and its category labels are conveyed to $\mathcal{D}_l$ to update the classification model. To implement this framework, we first introduce a classification model f_c to evaluate the classification performance with the annotated data. Here, we use a spectral–spatial residual network (SSRN) [41] architecture as the basic model, owing to its stability and high performance. The model is split into two components, i.e., feature embedding and sample classification.

For the feature embedding part, the backbone encodes an input $\boldsymbol{x}_i \in \mathbb{R}^n$ to a latent space of feature representations as

$$\boldsymbol{z}_i = f_b(\boldsymbol{x}_i, \boldsymbol{\theta}_{c1}) \tag{1}$$

where $\boldsymbol{z}_i \in \mathbb{R}^d$ is a d-dimensional feature formed by parameter $\boldsymbol{\theta}_{c1} \in \mathbb{R}^{n\times d}$.

For the sample classification part, after a fully connected layer and a softmax function, $\boldsymbol{z}_i$ is mapped to a class probability. This procedure can be written as

$$\hat{\boldsymbol{y}}_i = \text{softmax}(f_c(\boldsymbol{z}_i, \boldsymbol{\theta}_{c2})) \tag{2}$$

where $\hat{\boldsymbol{y}}_i \in \mathbb{R}^K$ is the set of output probabilities formed by the parameter $\boldsymbol{\theta}_{c2} \in \mathbb{R}^{d\times K}$. To optimize this objective, we minimize the cross-entropy loss $\mathcal{L}_c$ over the labeled set as follows:

$$\mathcal{L}_c(y, \hat{\boldsymbol{y}}) = -\frac{1}{N_l}\sum_{i=1}^{N_l}\sum_{k=1}^{K}\mathbb{I}(y_i)\log\big(\text{softmax}\big(\hat{\boldsymbol{y}}_i\big)\big) \tag{3}$$

where $\mathbb{I}(\cdot)$ is an indicator function reflecting the true label distribution. After model training is complete, we extract the feature representations of labeled and unlabeled samples with the parameter $\boldsymbol{\theta}_{c1}$ for the subsequent AL strategy.

B. Class-wise Graph Embedding-based AL

Unlike previous AL approaches, our sampling function is independent of the classification model and operates directly on the labeled and unlabeled data, allowing them to interact in a class-wise manner. First, we built binary graphs for each class using one-label and unlabeled data. Then, we feed the graphs into the CGCN to learn the relationship of labeled and unlabeled nodes. Finally, we apply class minimum uncertainty to query the most uncertain unlabeled samples.

1) Graph Construction: Based on true labels, we partition the labeled set $\mathcal{D}_l$ into K one-label sets, i.e., $\{\boldsymbol{c}_k\}_{k=1}^{K}$, each containing N_k labeled samples from the kth class. Then, we construct the one-label graph $\mathcal{G}$ for each category, consisting of nodes and edges. The nodes $\mathcal{V} \in \mathbb{R}^{(d\times N_{k+u})}$ are initialized with the one-label feature $f_b(\boldsymbol{c}_k) \in \mathbb{R}^{(d\times N_k)}$ and unlabeled feature $f_b(\mathcal{D}_u) \in \mathbb{R}^{(d\times N_u)}$, where N_{k+u} denotes the total number of samples in the nodes. The edges capture the relationships between nodes and represent their similarities. After performing ℓ_2 normalization on $\mathcal{V}$, the elements of the adjacency matrix $\mathcal{A}$ are initialized with vector products $\mathcal{S}$ (i.e., $\mathcal{S}_{ij} = \mathcal{V}_i^{\top}\mathcal{V}_j, \{i, j\} \in N_{k+u}$). The formula of $\mathcal{A}$ can be represented as

$$\mathcal{A} = D^{-1}(\mathcal{S} - \mathcal{I}) + \mathcal{I} \tag{4}$$

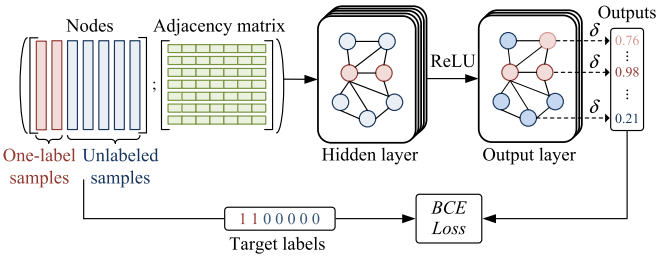


Fig. 3. Illustration of the CGCN architecture. Nodes and adjacency matrix, used as the input, are built from the representations of one-label samples and all unlabeled samples. The CGCN mainly consists of an input layer, a hidden layer, and an output layer. BCE loss is used to measure the difference between the predict scores and target labels.

where $\mathcal{A} \in \mathbb{R}^{(N_{k+u}\times N_{k+u})}$ is the resulting adjacency matrix. $\mathcal{I}$ is the identity matrix. D is the degree matrix of $\mathcal{S}$.

2) Network Structure: As Fig. 3 shows, we adopt a two-layer graph convolutional architecture. The first layer f_g^1, with parameters $\boldsymbol{\theta}_{g1} \in \mathbb{R}^{(d\times h)}$, can be summarized as

$$\mathcal{F} = f_g^1\big(\mathcal{V}, \mathcal{A}; \boldsymbol{\theta}_{g1}\big) \tag{5}$$

where $\mathcal{F} \in \mathbb{R}^{(h\times N_{k+u})}$ is a graph embedding, and h denotes the number of neurons in this layer. After the first layer, a rectified linear unit (ReLU) activation function is applied to maximize feature contribution. The second layer f_g^2 with parameters $\boldsymbol{\theta}_{g2} \in \mathbb{R}^{(h\times 1)}$ maps the nodes as the probabilities, which are then activated through a sigmoid function $\sigma(\cdot)$. This process can be defined as

$$\mathcal{P} = \sigma\big(f_g^2\big(\text{ReLU}(\mathcal{F}), \mathcal{A}; \boldsymbol{\theta}_{g2}\big)\big) \tag{6}$$

where $\mathcal{P} \in \mathbb{R}^{N_{k+u}}$ denotes the prediction probabilities of all the samples between 0 and 1. The probability close to 1 indicates that the sample is similar with the one-label set. Conversely, a lower score suggests high discrepancy regarding the sample. To optimize this objective, we use binary cross-entropy (BCE) loss [42] as follows:

$$\mathcal{L}_g(\mathcal{V}, \mathcal{A}) = -\frac{1}{N_k}\sum_{i=1}^{N_k}\log(\mathcal{P}(\boldsymbol{x}_i)) - \frac{\lambda}{N_u}\sum_{j=1}^{N_u}\log\big(1 - \mathcal{P}\big(\boldsymbol{x}_j\big)\big) \tag{7}$$

where the first term is applied to the labeled set, while the second term is used for the unlabeled set. Here, N_k and N_u represent the number of samples in the kth one-label and unlabeled sets, respectively. The parameter λ acts as a weighting factor to balance the contribution of two terms. To calculate binary losses, the labeled samples are taken as 1, and all the unlabeled samples (may from all the classes) are taken as 0. Note that this behavior does not harm the performance of binary classification. The reason is CGCN inherits the higher order feature representations from SSRN and can easily recognize the unlabeled samples in other classes. Moreover, the variable and uncertain samples with noise label may output lower probabilities, which provides an opportunity to better evaluate the intraclass uncertainty.

3) Uncertainty Sampling: To achieve this goal, we convert the predicted confidence score into uncertainty score. Suppose $\mathcal{P}_k(\boldsymbol{x})$ denotes the confidence of unlabeled samples to the kth

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Algorithm 1 The Proposed CGE-AL Strategy

Inputs: initial labeled set $\mathcal{D}_l$; unlabeled set $\mathcal{D}_u$; AL stages R; budget size B; number of classes K; uncertainty threshold c_m;
for $iter = 1$ *to* R **do**
  Train the classification model f_c using $\mathcal{D}_l$.
  Initialize the candidate set $\mathcal{D}_q$ as empty.
  for $k = 1$ *to* K **do**
    Select one-label set $\boldsymbol{c}_k$ from $\mathcal{D}_l$.
    Initialize nodes $\mathcal{V}$ by the representations of $\boldsymbol{c}_k$ and $\mathcal{D}_u$.
    Calculate $\mathcal{A}$ by (4).
    Train the CGCN using $\mathcal{V}$ and $\mathcal{A}$.
    Save the predicted uncertainty $\mathcal{U}_k(\mathcal{D}_u)$.
  end
  for $\boldsymbol{x}_u \in \mathcal{D}_u$ **do**
    if $\min\{\mathcal{U}_k(\boldsymbol{x}_u)\}_{k=1}^{K} \geq c_m$ **then**
      $\mathcal{D}_q = \mathcal{D}_q \cup \boldsymbol{x}_u$;
    end
  end
  Cluster the samples in $\mathcal{D}_q$ into B groups and select the closest samples $\mathcal{C}_q$ to each centroid.
  $\mathcal{D}_l = \mathcal{D}_l \cup \mathcal{C}_q$.
  $\mathcal{D}_u = \mathcal{D}_u - \mathcal{C}_q$.
end
Outputs: the optimal model f_c with $\mathcal{D}_l$

class, and the class uncertainty can be defined as follows:

$$\mathcal{U}_k(\boldsymbol{x}) = 1 - \mathcal{P}_k(\boldsymbol{x}) \tag{8}$$

where $\mathcal{U}_k(\boldsymbol{x}) \in \mathbb{R}^{N_u}$ denotes the degree to which the unlabeled samples are unknown for the labeled data in the kth class. After completing the CGCNs on all the graphs, we start to query the class minimum uncertainty of unlabeled samples. In the design of our sampling function, we expect that the queried samples are sufficiently different from the labeled samples of all the classes. If an unlabeled sample is unknown to all the labeled data, we assume that the classifier has trouble recognizing it correctly and its prediction result may be wrong and uncertain. This corresponds to the overall class uncertainty of the samples ought to be higher than other ordinary samples. To fulfill this condition, we compare the class minimum uncertainty with a predefined uncertainty threshold c_m. The uncertainty sampling function on unlabeled data can be written as

$$\mathcal{D}_q = \left\{\boldsymbol{x}_u \in \mathcal{D}_u | \min\{\mathcal{U}_k(\boldsymbol{x}_u)\}_{k=1}^{K} \geq c_m\right\} \tag{9}$$

where $\mathcal{D}_q$ represents the queried set. Note that c_m is set close to 0.5 to apply to the binary prediction results.

C. Diversity Selection and Classification

After selecting the uncertain unlabeled samples, the number of samples in the queried set $\mathcal{D}_q$ is probably larger than the budget size B. Note that we actively select B unlabeled samples in each iteration. Moreover, the majority of unlabeled samples from adjacent regions are likely to offer similar feature representations, which provides redundant information to the classifier. Therefore, we expect to choose B diverse samples from $\mathcal{D}_q$. To implement this goal, we cluster the samples in $\mathcal{D}_q$ into B groups using k-means on their feature representations. Subsequently, we select the closest samples to the centroid from each group as the candidate set $\mathcal{C}_q$. After human experts label them, the candidate set is added into the labeled set to update the classification model and the CGCN.

After all the AL stages are finalized, we use an individual test set $\mathcal{D}_t$ to assess the classification effectiveness of our classification model f_c with the optimal labeled set.

IV. Experiments

A. Datasets and Experimental Settings

1) Datasets' Description: We evaluate the proposed method and the compared methods on three public HSI datasets: Indian Pines, Pavia University, and Houston 2013. These datasets are described below.

1) The Indian Pines dataset was captured by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor over Northwestern Indiana in 1992. The dataset consists of 145 × 145 pixels and 200 spectral bands, with a spatial resolution of 20 m per pixel (24 water absorption channels were removed). As the scene was captured in June, some crops (such as corn and soybeans) are still in the early stage of growth. The class names and the number of labeled and reference samples are given in Table I.
2) The University of Pavia image was collected by the Reflective Optics System Imaging Spectrometer-3 (ROSIS-3) sensor over the University of Pavia and its surroundings in Pavia, Italy. The image comprises 610 × 340 pixels, with a spatial resolution of 1.3 m per pixel and 115 bands. After removing 12 noisy bands, the experiments were conducted on the remaining 103 bands. A fixed number of samples for labeled and reference were used as described in Table I.
3) The Houston 2013 scene was acquired by the ITRES CASI-1500 sensor over the campus of the University of Houston and its neighboring rural areas, and it was used in the 2013 GRSS Data Fusion Contest. The data comprises 144 spectral bands in the 380–1050-nm region, and 349 × 1905 pixels with a spatial resolution of 2.5 m. This image contains 15 classes, where the numbers of labeled and reference samples are detailed in Table I.

2) Evaluation Metrics: Three widely used evaluation metrics—overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ)—are adopted to quantify the classification performance. These metrics are calculated by averaging the results achieved after ten repeated experiments.

3) AL Experimental Settings: Initially, we randomly select ten labeled samples from the reference data for each class as an initial labeled set, and 4000 samples are selected as an unlabeled set for all the datasets. The remaining samples are used as the test set to evaluate the performance of the model. The test sets contain 6089, 38 686, and 10 879 samples on the Indian Pines, Pavia University, and Houston 2013, respectively.

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TABLE I
CATEGORY NAME, SAMPLES' NUMBERS OF INITIAL LABELED SET, AND ALL REFERENCE SETS OF EACH CLASS ON THE INDIAN PINES, PAVIA UNIVERSITY, AND HOUSTON 2013 DATASETS

Indian Pines				Pavia University			Houston 2013		
No.	Category	Labeled	Reference	Category	Labeled	Reference	Category	Labeled	Reference
1	Alfalfa	10	46	Asphalt	10	6631	Healthy Grass	10	1251
2	Corn Notill	10	1428	Meadows	10	18649	Stressed Grass	10	1254
3	Corn Mintill	10	830	Gravel	10	2099	Synthetis Grass	10	697
4	Corn	10	237	Trees	10	3064	Tree	10	1244
5	Grass Mowed	10	483	Painted Meatal Sheets	10	1345	Soil	10	1242
6	Grass Trees	10	730	Bare Soil	10	5029	Water	10	325
7	Grass Pasture	10	28	Bitumen	10	1330	Residential	10	1268
8	Hay Windrowed	10	487	Self Blocking Bricks	10	3682	Commercial	10	1244
9	Oats	10	20	Shadows	10	947	Road	10	1252
10	Soybean Notill	10	972				Highway	10	1227
11	Soybean Mintill	10	2455				Railway	10	1235
12	Soybean Clean	10	593				Parking Lot1	10	1233
13	Wheat	10	205				Parking Lot2	10	469
14	Woods	10	1265				Tennis Court	10	428
15	Buildings	10	386				Running Track	10	660
16	Stone	10	93						
	Total	160	10249	Total	90	42776	Total	150	15029

The total number of iterations is five (i.e., AL stages $R = 5$) in our AL experiments. In each iteration, we actively select 30, 20, and 30 unlabeled samples (i.e., budget size B) on the Indian Pines, Pavia University, and Houston 2013 datasets, respectively. In k-means clustering, the uncertain candidates are grouped into B clusters.

4) Competitive Approaches: We compare the proposed method with three state-of-the-art AL approaches and six SSRN-based sampling techniques.

1) CNN with Markov random field (CNN-MRF) [43] integrates 2-D CNN and BvSB to select the most informative samples.
2) 3D-CNN [44] adopts the 3-D CNN architecture and mutual information criteria to actively select the heterogeneous samples.
3) FAAL [28] exploits a GAN to identify whether unlabeled samples are labeled or not in a latent feature space.
4) Random selects samples from the unlabeled pool in a random fashion and is used as a baseline for measuring the classification accuracy.
5) BvSB computes the interval between the prediction probabilities of the best and second best classes, and then selects the unlabeled samples with the lowest interval.
6) Entropy considers the unlabeled samples with highest entropy to be rich in information.
7) Learning loss (LL) [45] predicts the target loss of unlabeled data and selects the samples with wrong predictions for labeling.
8) CoreGCN [38] learns the GCN with the CoreSet technique to measure the uncertainty of unlabeled samples.

B. Implementation Details

For the feature embedding network, we used SSRN for the classification task, where HSI patches of size 11×11 pixels are fed to the model with a batch size of 128. The objective function is the cross-entropy loss, and we used stochastic gradient descent (SGD) as the optimizer, with a learning rate of 5×10^{-4} for 120 epochs. The feature embeddings with the trained parameters from SSRN were passed to the CGCN.

The CGCN module comprises two graph convolutional layers, a dropout layer, and a linear layer. The hidden and output layers have 32 and one neurons, respectively. To represent the uncertainty of samples, the output score was compressed to a range between 0 and 1 using the sigmoid function. We optimized the objective using BCE loss and used the Adam optimizer with a learning rate of 1×10^{-3} for CGCN optimization. The parameter λ is set to 1.2 based on the optimal results.

All the experiments are performed on a Windows PC system with an Intel Core i7-7800X CPU, 32-GB RAM, and an NVIDIA GTX 1080Ti 11-GB GPU. We implemented all the SSRN-based sampling techniques on the PyTorch platform and FAAL on the TensorFlow platform. We implemented CNN-MRF and 3D-CNN on MATLAB R2020b.

C. Sensitivity Analysis

1) Parameter Sensitivity: First, we analyze the impact of the number of epochs in the CGCN. Fig. 4(a) illustrates the classification performance obtained using different numbers of epochs with the Indian Pines dataset. The results demonstrate that 100 epochs is enough for CGCN to learn the relationships between labeled and unlabeled samples. Therefore, 100 epochs are selected as an adequate setting for CGCN on the used datasets. Then, an experiment was conducted to analyze the impact of the uncertainty threshold c_m in the range $\{0.525, 0.55, 0.6, 0.7\}$. The classification accuracies are given in Fig. 4(b). The results obtained using a value of 0.05 for this parameter are stable and hold a small variance in all the AL stages. Based on this, we set the sampling threshold to 0.55 for all the datasets.

2) Ablation Study: In addition, we perform ablation studies on the acquisition function for the Pavia University dataset. In Fig. 4(c), cluster and CGCN, as two main components of the proposed method (i.e., CGE-AL 2-Layer), were used to select the unlabeled samples for labeling, respectively. The results of them are better than the baseline, but lower than the proposed method. Moreover, a one-layer CGCN structure is considered in our acquisition function, which means that the

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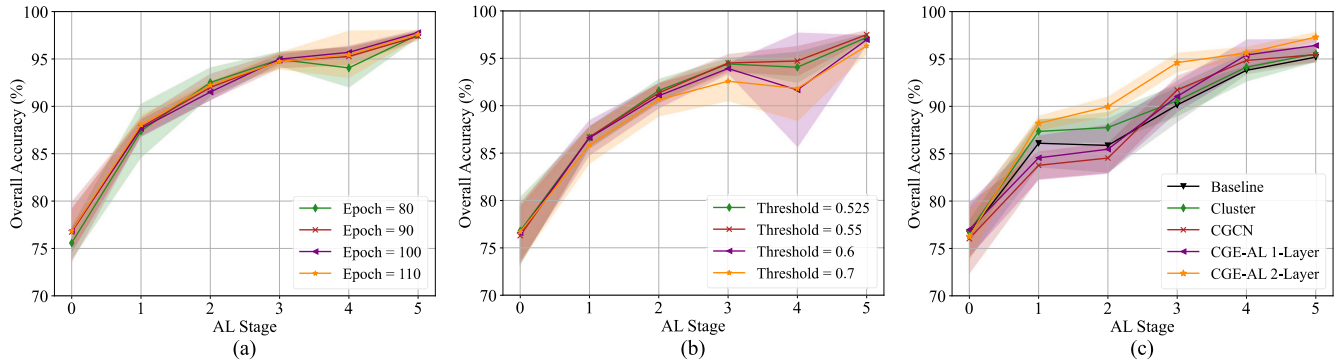


Fig. 4. Sensitivity analysis of the acquisition function in the proposed method. (a) Epochs of the CGCN. (b) Uncertain threshold c_m. (c) Ablation study with various structures. (CGE-AL 2-Layer is the proposed method.)

TABLE II

CLASSIFICATION ACCURACY COMPARISONS OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASS FOR THE INDIAN PINES DATASET WITH TOTAL 310 SAMPLES FOR TRAINING. THE BEST RESULTS ARE IN BOLD

	Conventional AL Methods			SSRN-based Sampling Techniques					
No.	CNN-MRF	3D-CNN	FAAL	Random	BvSB	Entropy	LL	CoreGCN	CGE-AL
1	**100.0**	97.83	88.08	87.50	**100.0**	95.45	93.42	98.48	**100.0**
2	94.09	83.33	73.44	84.53	96.21	89.21	78.10	94.56	**99.29**
3	89.26	51.69	88.71	90.04	**99.37**	94.32	80.71	93.97	97.14
4	97.56	47.26	87.73	91.22	87.92	87.97	75.55	**98.17**	93.15
5	96.47	66.87	83.87	97.35	96.80	90.78	90.18	**98.43**	97.79
6	96.93	**100.0**	97.84	94.51	99.76	96.77	96.81	99.45	99.76
7	**100.0**	**100.0**	**100.0**	34.38	**100.0**	85.71	85.53	**100.0**	**100.0**
8	99.46	99.79	**100.0**	**100.0**	99.64	**100.0**	99.62	**100.0**	**100.0**
9	**100.0**	**100.0**	**100.0**	57.89	52.38	**100.0**	79.22	88.43	**100.0**
10	85.77	65.53	86.18	79.39	91.03	86.28	81.84	90.07	**98.06**
11	**98.88**	84.56	94.37	93.50	98.39	84.92	76.04	94.43	97.89
12	90.77	78.08	77.56	94.88	92.74	88.13	76.27	**97.75**	96.52
13	**100.0**	**100.0**	97.39	**100.0**	97.52	98.32	91.84	98.79	96.43
14	96.07	86.72	97.55	98.83	99.87	99.34	95.47	**99.96**	99.08
15	**99.46**	89.90	89.53	94.09	92.86	87.55	85.62	97.47	95.42
16	**100.0**	**100.0**	95.88	88.24	93.75	91.84	84.52	90.20	84.75
OA (%)	95.17±0.25	80.71±1.03	89.35±2.10	91.25±0.26	96.64±1.08	90.63±0.49	82.38±3.18	95.87±0.78	**97.98±0.20**
AA (%)	96.55±0.19	84.47±1.97	91.13±0.87	86.65±0.52	93.64±2.06	92.29±0.40	85.67±3.74	96.26±0.71	**97.20±0.76**
$\kappa \times 100$	94.48±0.28	78.07±1.01	87.82±2.41	90.01±0.29	96.17±1.23	89.25±0.57	79.87±3.57	95.27±0.89	**97.69±0.23**

nodes are projected directly to the result without going through a hidden layer. This structure yielded a promised result but also lower than those on the two-layer CGCN structure.

3) Embedding Backbone: To further evaluate the effectiveness of the studied sampling strategies, we analyze the performance of those techniques with basic DL-based embedding backbones on the Pavia University dataset. Fig. 5 illustrates the classification accuracies with several backbones, i.e., MLP, 2D-CNN, 3D-CNN, ViT, and SSRN. In Fig. 5, BvSB achieves better performance in comparison to other sampling strategies by only querying the best and second best classes of samples. Except for the ViT architecture, the proposed method outperforms other competitors in all the backbones. Besides, the results of ResNet are higher than those of other backbones, which means that this architecture may be an optimal choice to serve as the embedding backbone in our AL framework for HSIs. In conclusion, the experimental results prove the robustness of the proposed class-wise uncertainty sampling strategy in various backbone networks.

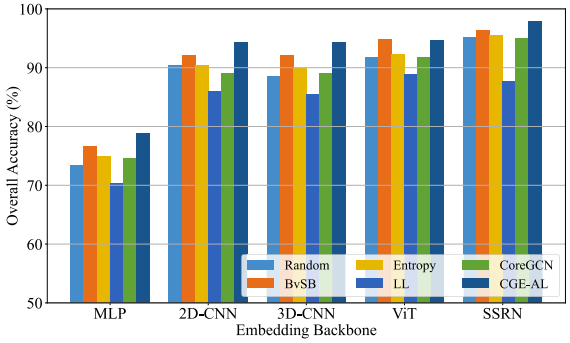


Fig. 5. Qualitative performance comparison of various sampling techniques with different embedding backbones.

D. Prediction Result Comparison

A series of experiments have been carried out to evaluate the performance of the compared AL methods and SSRN-based sampling techniques. Tables II–IV and Figs. 6–8 illustrate the obtained classification accuracies and maps for the Indian Pines, Pavia University, and Houston 2013 datasets, respectively.

1) Indian Pines: The first experiment was conducted on the Indian Pines dataset using 160 initial labeled samples and 150 selected samples across five AL rounds. As Table II shows, 3D-CNN and LL performed worse than other com-

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TABLE III

CLASSIFICATION ACCURACY COMPARISONS OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASS FOR THE PAVIA UNIVERSITY DATASET WITH TOTAL 190 SAMPLES FOR TRAINING. THE BEST RESULTS ARE IN BOLD

	Conventional AL Methods			SSRN-based Sampling Techniques					
No.	CNN-MRF	3D-CNN	FAAL	Random	BvSB	Entropy	LL	CoreGCN	CGE-AL
1	96.16	83.22	85.79	96.53	98.10	95.54	91.44	97.13	**98.79**
2	99.29	90.48	98.87	97.45	99.00	97.26	96.75	98.48	**99.61**
3	81.01	63.81	89.56	86.00	90.75	87.17	81.96	92.47	**99.51**
4	95.26	80.05	91.68	96.07	97.63	98.20	85.28	87.49	**98.85**
5	98.54	**99.94**	98.39	99.59	**99.94**	99.69	98.64	99.20	99.92
6	**99.97**	85.34	96.53	96.27	96.43	96.30	71.48	95.48	98.80
7	95.82	70.14	**99.84**	89.13	98.14	89.88	83.24	92.16	97.42
8	**95.48**	77.10	83.46	87.15	91.74	89.28	84.47	84.28	86.54
9	91.45	85.15	84.99	97.58	99.37	98.52	89.72	98.02	**100.0**
OA (%)	97.07±0.39	85.09±3.27	93.98±3.77	95.19±0.75	97.39±0.42	95.54±0.97	87.60±3.83	95.01±0.51	**97.99±0.42**
AA (%)	94.78±1.25	81.69±5.37	92.12±2.86	93.97±1.61	96.79±0.56	94.65±2.00	87.00±3.50	93.86±0.99	**97.72±0.28**
$\kappa \times 100$	96.12±0.52	80.43±4.06	92.08±4.89	93.62±0.99	96.53±0.55	94.07±1.28	83.98±4.68	93.41±0.65	**97.33±0.56**

TABLE IV

CLASSIFICATION ACCURACY COMPARISONS OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASS FOR THE HOUSTON 2013 DATASET WITH TOTAL 300 SAMPLES FOR TRAINING. THE BEST RESULTS ARE IN BOLD

	Conventional AL Methods			SSRN-based Sampling Techniques					
No.	CNN-MRF	3D-CNN	FAAL	Random	BvSB	Entropy	LL	CoreGCN	CGE-AL
1	94.32	79.28	83.40	**98.67**	96.78	98.59	90.55	97.41	98.58
2	**98.39**	89.81	97.46	92.54	95.49	91.19	83.72	96.20	95.76
3	98.47	99.60	97.19	99.86	**100.0**	99.78	99.72	97.14	**100.0**
4	96.22	81.53	88.62	96.25	**99.05**	91.46	88.76	92.54	97.62
5	99.67	99.61	99.91	95.01	**99.96**	95.77	92.31	98.25	96.96
6	93.93	89.29	94.31	**100.0**	**100.0**	96.35	82.26	99.84	**100.0**
7	88.62	66.85	69.49	87.08	94.61	91.35	71.30	88.83	**98.90**
8	88.41	65.88	74.70	96.09	93.64	93.31	78.63	95.97	**98.66**
9	**88.87**	69.09	63.28	84.99	88.41	88.08	85.14	80.61	86.73
10	93.33	52.08	89.97	84.48	92.62	90.39	84.63	86.98	**97.29**
11	88.47	69.65	86.22	93.60	96.69	91.73	82.14	90.87	**99.27**
12	80.20	71.61	87.87	88.09	91.15	87.55	86.17	85.91	**94.21**
13	91.64	50.75	82.91	89.39	**98.42**	92.59	84.38	83.25	96.47
14	98.21	99.67	**100.0**	96.51	98.77	91.74	82.80	99.00	99.89
15	98.07	99.30	**100.0**	96.51	97.53	96.80	96.99	96.64	97.23
OA (%)	92.51±1.23	77.11±3.48	85.96±1.40	92.20±0.95	95.38±0.44	92.42±0.82	84.50±1.85	91.58±2.25	**96.62±0.39**
AA (%)	93.12±1.03	78.93±3.59	87.69±1.61	96.09±0.92	96.20±0.26	93.11±0.27	85.97±2.76	92.63±1.86	**97.17±0.50**
$\kappa \times 100$	91.90±1.33	75.25±3.76	84.82±1.51	91.57±1.03	95.01±0.48	91.80±0.88	83.25±1.98	90.90±2.43	**96.35±0.42**

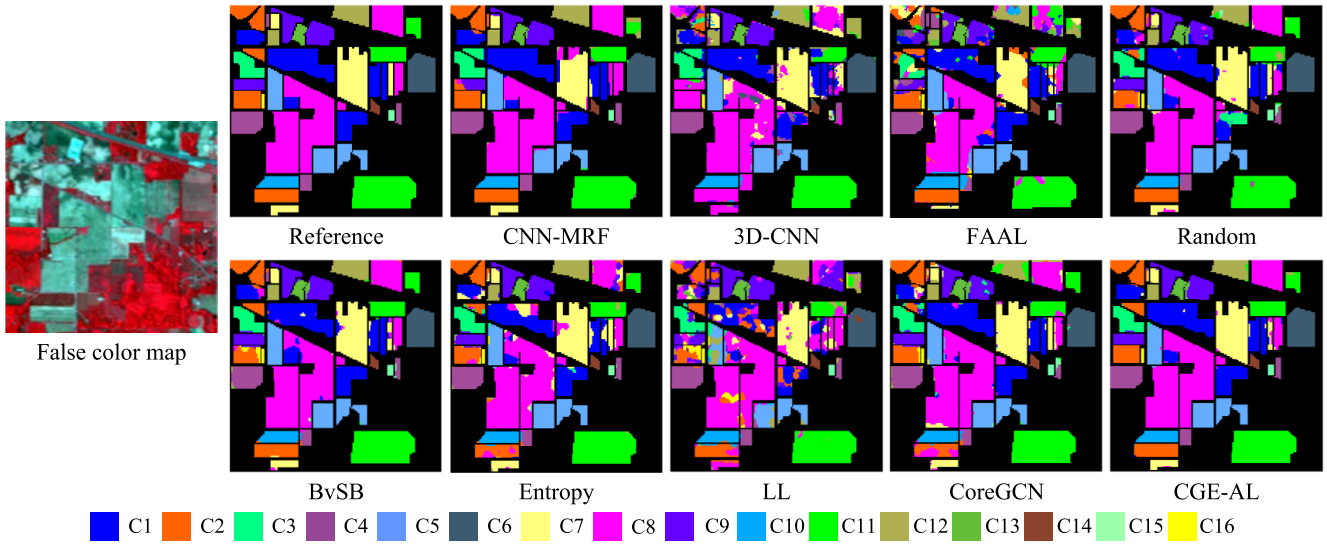


Fig. 6. Classification map comparisons by the compared methods for the Indian Pines dataset.

pared methods. Random served as the baseline for the SSRN backbone and achieved an OA of 91.25%. Entropy considered merely the information content of unlabeled data, but disregarded the distribution of the current labeled set, resulting in reduced effectiveness for newly labeled samples. CoreGCN gained better performances in comparison to other methods

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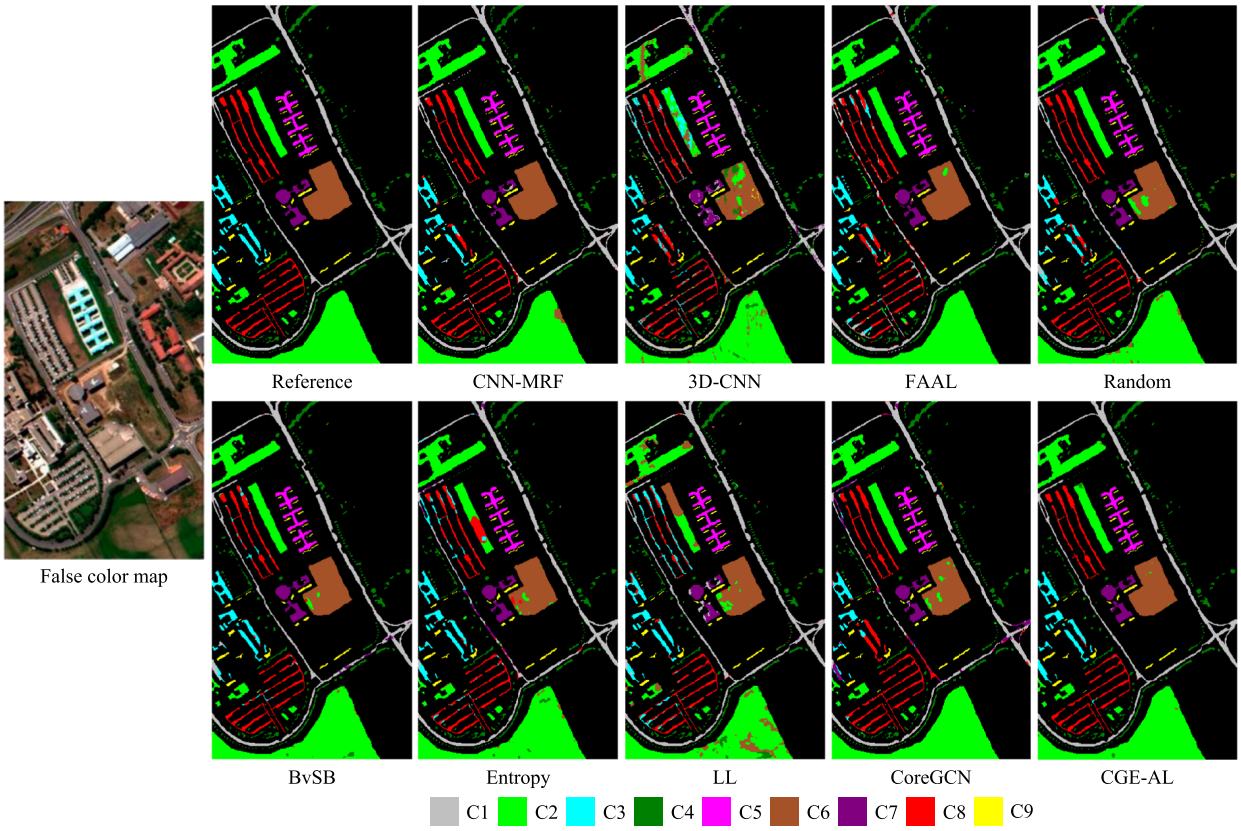


Fig. 7. Classification map comparisons of the compared methods for the Pavia University dataset.

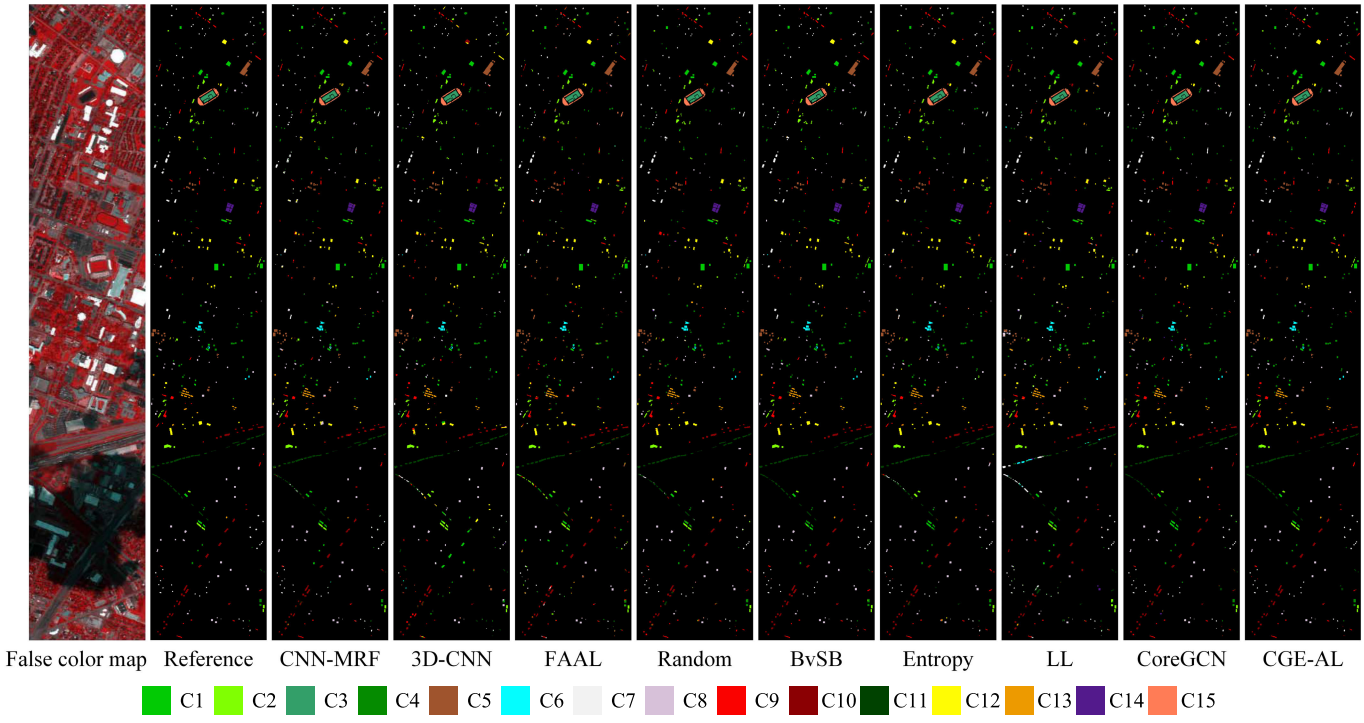


Fig. 8. Comparison of classification results by the compared methods for the Houston 2013 dataset.

by measuring the relationships between labeled and unlabeled samples. CNN-MRF and BvSB yielded better performance than that of other sampling techniques. On the other hand, CGE-AL delivered the best results among the studied methods and attained a significant improvement compared with the baseline, achieving an OA improvement of 6.73%. Fig. 6 illustrates the obtained classification maps. The maps of 3D-CNN, FAAL, and LL contain a high amount of noise. CoreGCN suffered from misclassification errors on the complex region (i.e., the top left of the image). The visual results of random

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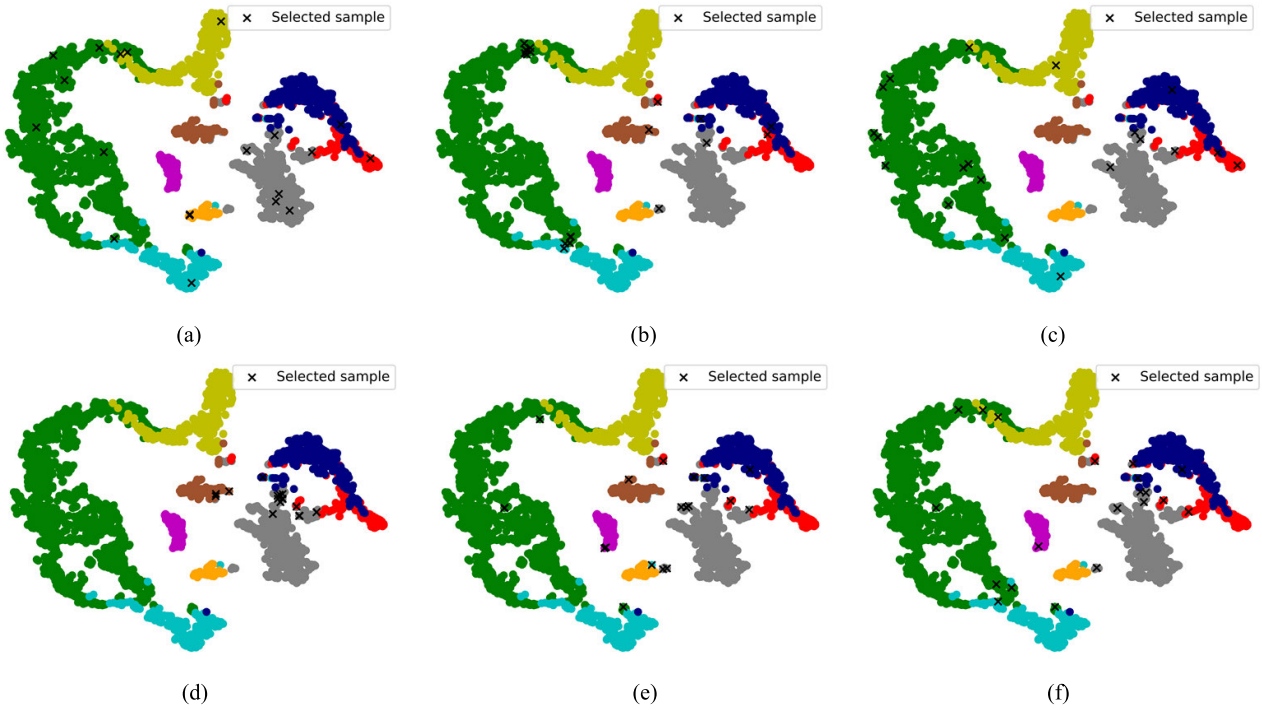


Fig. 9. T-SNE visualization maps of the studied sampling techniques in the latent feature space learned by SSRN. (a) Random. (b) BvSB. (c) Entropy. (d) LL. (e) CoreGCN. (f) CGE-AL.

and entropy are better than the previous methods, but with some edge noise. BvSB and entropy delivered better classification results, with smooth regions, although with some noise at the boundary between the Alfalfa and Hay Windrowed (i.e., C1 and C8). CNN-MRF formed a clear and smoothed visualization map expect for a few errors in detail. In contrast, CGE-AL generated a high-quality map with well-defined edges and textures, which benefited from robust labeled data.

2) Pavia University: We performed the second experiment on the Pavia University dataset with 90 initial labeled samples and 100 actively selected samples in five iterations. Table III and Fig. 7 illustrate the classification accuracies and maps obtained by the compared methods with this dataset. The classification results of 3D-CNN, FAAL, and LL are still worse than random. Entropy and CoreGCN provided similar results compared with the baseline, which means that the selected samples are not useful in improving the performance of the model. CNN-MRF and BvSB achieved better results than other methods in terms of the obtained accuracies. Furthermore, CGE-AL provided and labeled a set of high-quality data, effectively improving the performance of the model. We observed a similar tendency in Fig. 7. 3D-CNN and LL obtained classification maps with significant noise. The maps of entropy are smooth over the large regions, but there is still some noise. CNN-MRF, FAAL, and CoreGCN showed better results in the Meadows area, despite some noise being present in the buildings area. Conversely, CGE-AL provided a clear classification map that is closer to the reference map.

3) Houston 2013: The last experiment was conducted with the Houston 2013 dataset using 150 initial samples and 150 actively selected samples. Table IV and Fig. 8 report the quantitative and qualitative results, respectively. In Table IV, the results of FAAL outperformed those of 3D-CNN and LL in terms of accuracies, but were inferior to those provided by other AL methods. In comparison to random, the annotated data selected by CoreGCN were useless for improving the performance of the model. Entropy and BvSB obtained limited improvements. CGE-AL achieved a 4.42% OA improvement over the baseline and outperformed other AL methods. As illustrated in Fig. 8, 3D-CNN and LL produced classification errors in the region of road (i.e., C9). The visualization maps of BvSB and entropy are better than those of other methods, with varying degrees of noise estimation in the obtained maps. Furthermore, CNN-MRF and CoreGCN and CoreGCN outperformed the above methods. CGE-AL is able to provide a closer visual result with regard to the ground truth.

E. Effectiveness Verification

1) Sampling Behavior: To further analyze the sampling technique of the proposed method, we performed a qualitative comparison of five sampling techniques on the Pavia University dataset. Fig. 9 presents the t-SNE visualization maps of the compared methods on the same latent feature space. Entropy selects the most representative samples but ignored their uncertainty. BvSB and CoreGCN are only concerned with the uncertainty and query the samples on the class boundaries. In contrast, CGE-AL selects samples at multiclass boundaries while preserving a sparse distribution to ensure a good representativeness of the unlabeled set.

2) Labeled Data Quantity: Fig. 10 illustrates the classification performance of the studied sampling methods with various numbers of labeled data for the three considered HSI datasets.

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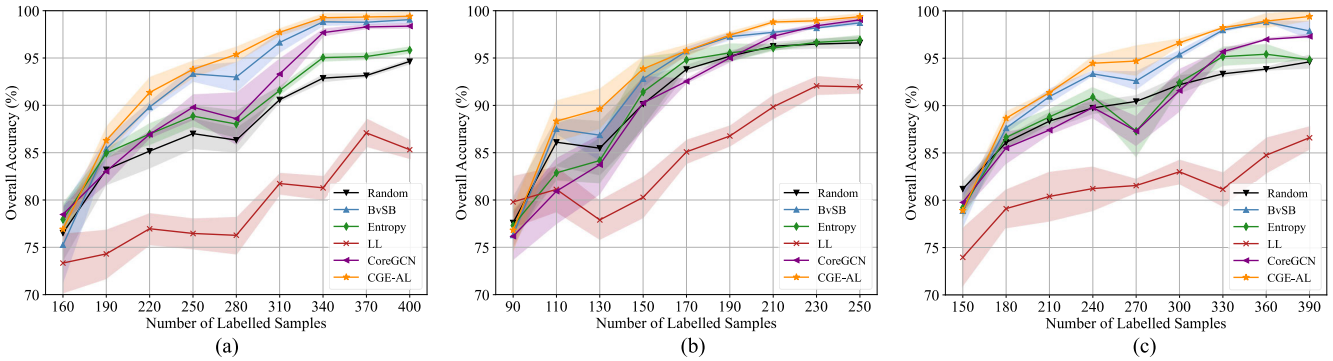


Fig. 10. Qualitative performance comparison of various sampling techniques with different numbers of labeled data on three used datasets. (a) Indian Pines. (b) Pavia University. (c) Houston 2013.

TABLE V

QUALITATIVE PERFORMANCE COMPARISON OF THE COMPARISON METHODS WITH ALL UNLABELED DATA

AL Method	OA(%)	AA(%)	$\kappa \times 100$
CNN-MRF	94.65±1.05	96.11±0.98	93.87±1.21
3D-CNN	80.09±1.59	86.26±2.30	77.27±1.96
FAAL	84.17±3.01	88.61±2.15	82.06±3.38
Random	91.14±0.27	90.41±0.63	89.89±0.29
BvSB	96.73±0.89	95.77±1.45	96.26±1.02
Entropy	91.40±0.85	89.38±1.42	90.17±0.98
LL	79.17±3.33	80.46±3.18	76.36±3.81
CoreGCN	95.19±0.90	96.65±0.62	94.52±1.03
CGE-AL	**97.60±0.42**	**97.78±1.37**	**97.25±0.48**

TABLE VI

TRAINING TIME ($t_t \times 10^2$) AND SAMPLING TIME ($t_s \times 10^2$) IN SECONDS OF THE COMPARISON METHODS

AL Method	Indian Pines		Pavia University		Houston 2013	
	t_t	t_s	t_t	t_s	t_t	t_s
CNN-MRF	0.62	0.01	2.26	0.01	0.96	0.01
3D-CNN	1.76	0.01	5.12	0.01	0.95	0.01
FAAL	0.40	3.49	0.28	3.37	0.36	3.36
Random	2.01	0.06	0.63	0.03	1.48	0.02
BvSB	2.02	0.11	0.63	0.06	1.45	0.04
Entropy	2.04	0.12	0.64	0.06	1.48	0.04
LL	2.12	0.31	0.98	0.33	1.83	0.21
CoreGCN	2.08	0.21	0.63	0.12	1.50	0.11
CGE-AL	2.12	0.78	0.69	0.74	1.48	0.78

The shaded area indicates the standard variance of the OAs in five repeated experiments. Fig. 10(a) shows the performance for the Indian Pines dataset. The results of CGE-AL consistently outperform those achieved by other sampling methods in each stage. The same conclusion can be extracted from the experiments with other datasets. Specifically, we can observe in Fig. 10(b) and (c) that after selecting 190 and 300 labeled samples, the proposed method achieves 97.42% and 96.63% accuracy on the Pavia University and Houston 2013 datasets. The CGE-AL method achieves the most satisfactory classification results with respect to other competitive methods. It is demonstrated that class minimum uncertainty sampling offers a set of informative and representative data, which is able to heavily promote the performance and robustness of the classification model.

3) All Unlabeled Data: To avoid leaving out the valuable samples in the preselection of the unlabeled set, we conduct an experiment on the Indian Pines dataset using all the reference data except the labeled data as the unlabeled set. After completing the whole AL process, the unselected unlabeled data (i.e., 9939 samples) are used to evaluate the performance of the model with the current labeled set. In this case, the models must sampling valuable samples on a larger unlabeled set and classify a more complex test set. In Table V, CGE-AL still achieved the best results than other compared methods owing to its powerful sampling function.

4) Computational Cost: We investigate the total computational cost of the proposed method and CGE-AL in five rounds for the used datasets. Table VI presents a comparison of their training time ($t_t \times 10^2$) and sampling time ($t_s \times 10^2$) in seconds. CNN-MRF and 3D-CNN are very fast because their

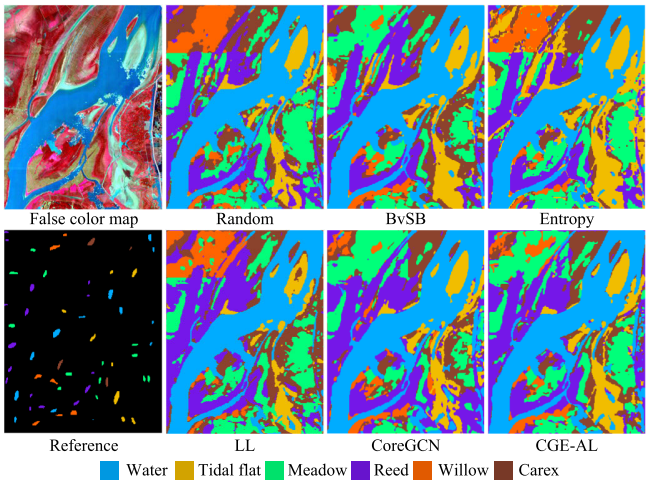


Fig. 11. Illustration comparison of the studied sampling techniques. The OAs of them are as follows: random (94.28 ± 2.10%), BvSB (96.32 ± 1.84%), entropy (94.26 ± 2.30%), LL (91.15 ± 2.56%), CoreGCN (98.42 ± 1.12%), and CGE-AL (99.24 ± 1.08%).

sampling functions are readily implemented without training on the MATLAB platform. CGE-AL incurs in an acceptable computational cost. In contrast, FAAL also spent a certain amount of time training the GAN to discriminate the unlabeled samples.

V. APPLICATION TO A SATELLITE DATASET

To further verify the robustness of our sampling method, we conduct experiments on Gaofen-5 (GF-5) satellite data. The used wetland scene was captured by the GF-5 satellite over Xinqiang River and its surroundings in Yueyang, Hunan,

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China. The image comprised 456 × 352 pixes and 310 bands. We manually labeled 4816 samples for six categories in this scene. For wetland scenes, it is difficult to visit the area in the field, which makes manual labeling extremely challenging. As a result, we consider that it makes sense to incorporate AL methods into the wetland mapping process. In this experiment, we used 18 labeled samples and actively selected 25 samples in five stages. Fig. 11 illustrates the obtained classification maps obtained by different methods. Compared with random, entropy and LL do not improve the classification results for this dataset. On the contrary, although water and land are intertwined, CGE-AL effectively separates the water and land. Moreover, the results of CGE-AL outperform those of other sampling methods in terms of accuracy and stability.

VI. Conclusion and Future Work

To alleviate the data annotation demand for the DL-based models, we propose a new class minimum uncertainty-based AL approach constructed on the CGE-AL framework. First, with a small number of labeled data, the classifier is trained to reference latent features of labeled and unlabeled samples. Then, we construct one-label graphs for each category among one-label and unlabeled data. After that, CGCNs are executed on these graphs to evaluate the uncertainty of the unlabeled nodes to all the classes. Finally, we adopt class minimum uncertainty and k-means to query the candidates from the unlabeled set. The candidates are labeled and added to the labeled set to update the parameters of our classifier and CGCNs. Our experimental results with four HSI datasets demonstrate excellent performance of the proposed method in comparison to other AL approaches. In future, we will extend the proposed AL method to the cross-modal classification task to promote the utilization of multimodal remote sensing data. Also, we will consider the concept of self-supervised learning and investigate how to better exploit large amounts of unlabeled data to mine some interesting and valuable information from them.

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Xiaolong Liao (Graduate Student Member, IEEE) received the B.S. degree in automation from the Hunan Institute of Science and Technology, Yueyang, China, in 2021, where he is currently pursuing the M.S. degree in information and communication engineering.

His research interests include hyperspectral image processing and remote sensing image registration.



Bing Tu (Senior Member, IEEE) received the M.S. degree in control science and engineering from the Guilin University of Technology, Guilin, China, in 2009, and the Ph.D. degree in mechatronic engineering from the Beijing University of Technology, Beijing, China, in 2013.

From 2015 to 2016, he was a Visiting Researcher with the Department of Computer Science and Engineering, University of Nevada, Reno, NV, USA, which is supported by the China Scholarship Council. He is currently a Full Professor with the Nanjing University of Information Science and Technology, Nanjing, China. His research interests include sparse representation, pattern recognition, and analysis in remote sensing.

Dr. Tu is an Associate Editor of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.



Jun Li (Fellow, IEEE) received the B.S. degree in geographic information systems from Hunan Normal University, Changsha, China, in 2004, the M.E. degree in remote sensing from Peking University, Beijing, China, in 2007, and the Ph.D. degree in electrical engineering from the Instituto de Telecomunicações, Instituto Superior Técnico (IST), Universidade Técnico de Lisboa, Lisbon, Portugal, in 2011.

From 2013 to 2021, she was a Full Professor with Sun Yat-sen University, Guangzhou, China. Since 2022, she has been with the China University of Geosciences, Wuhan, China, as a Full Professor. She has received several prestigious funding grants at the national and international levels. She has authored more than 160 journal citation report (JCR) articles, 60 international conference papers, and a book chapter.

Dr. Li has been serving as the Editor-in-Chief for the *Journal of Selected Topics in Applied Earth Observations and Remote Sensing* since 2021.



Antonio Plaza (Fellow, IEEE) received the M.Sc. and Ph.D. degrees in computer engineering from the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, University of Extremadura, Cáceres, Spain, in 1999 and 2002, respectively.

He is currently the Head of the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, University of Extremadura. He has authored more than 600 publications, including more than 200 JCR journal articles (more than 160 in IEEE journals), 23 book chapters, and around 300 peer-reviewed conference proceeding papers. His research interests include hyperspectral data processing and parallel computing of remote sensing data.

Dr. Plaza was a member of the Editorial Board of the IEEE Geoscience and Remote Sensing Newsletter from 2011 to 2012 and the *IEEE Geoscience and Remote Sensing Magazine* in 2013. He was also a member of the Steering Committee of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS). He is a fellow of IEEE for contributions to hyperspectral data processing and parallel computing of Earth observation data. He was a recipient of the recognition of Best Reviewers of the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS in 2009 and the Best Reviewer of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING in 2010, for which he has served as an Associate Editor from 2007 to 2012. He was also a recipient of the Best Column Award of the *IEEE Signal Processing Magazine* in 2015, the 2013 Best Paper Award of the JSTARS journal, and the Most Highly Cited Paper (2005–2010) in the *Journal of Parallel and Distributed Computing*. He received best paper awards at the IEEE International Conference on Space Technology and the IEEE Symposium on Signal Processing and Information Technology. He has served as the Director of Education Activities for the IEEE Geoscience and Remote Sensing Society (GRSS) from 2011 to 2012 and the President of the Spanish Chapter of IEEE GRSS from 2012 to 2016. He has reviewed more than 500 manuscripts for more than 50 different journals. He has served as the Editor-in-Chief for the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING from 2013 to 2017. He has guest edited ten special issues on hyperspectral remote sensing for different journals. He is also an Associate Editor of the IEEE ACCESS (receiving the recognition as an Outstanding Associate Editor of the journal in 2017). Additional information is available at http://www.umbc.edu/rssipl/people/aplaza.

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